

Multi-criteria ABC: dollar usage, scarcity and quality problems

Part 4 · How do we begin? — solution report with data, method, results and charts

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1. Part 4 · How do we begin? — Multi-criteria ABC: dollar usage, scarcity and quality problems

1. Multi-criteria ABC: dollar usage, scarcity and quality problems

Weights: dollar usage **0.6**, scarcity **0.25**, quality **0.15**. Both classifications use the 20/30/50 % items rule so that they are comparable.

154 (77%) Items keeping their class 0 C → A upgrades	23 Items upgraded	23 Items downgraded
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Adding scarcity and quality to the ABC criteria

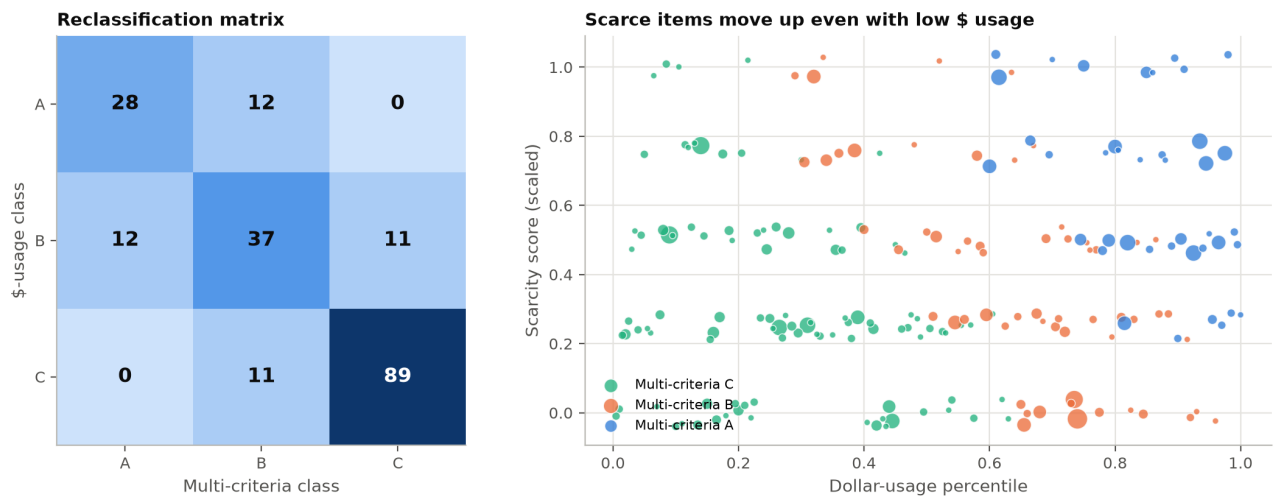


Figure 1.1 — Left: counts moving between classes. Right: bubble size $\propto$ quality incidents.

Largest upgrades (items that deserve more control than \$ usage alone suggests).

sku	category	annual dollar usage	scarcity score 1to5	quality incidents per year	score	class dollar	class multi
SKU-0167	Finished good	\$7,876	5	4	0.828	B	A
SKU-0155	Packaging	\$9,548	4	6	0.805	B	A
SKU-0034	Sub-assembly	\$3,821	5	8	0.764	B	A
SKU-0186	Sub-assembly	\$9,489	3	5	0.731	B	A
SKU-0175	Finished good	\$3,621	5	2	0.721	B	A
SKU-0070	Purchased part	\$4,892	4	3	0.706	B	A
SKU-0086	Sub-assembly	\$7,617	3	4	0.700	B	A
SKU-0017	Purchased part	\$9,184	3	2	0.698	B	A
SKU-0126	Raw material	\$5,618	5	0	0.695	B	A
SKU-0023	Packaging	\$3,339	4	6	0.685	B	A
SKU-0106	Raw material	\$9,417	4	0	0.684	B	A
SKU-0077	Raw material	\$5,541	4	1	0.677	B	A
SKU-0108	MRO supply	\$1,378	5	6	0.579	C	B
SKU-0145	Finished good	\$1,693	4	6	0.556	C	B
SKU-0149	Raw material	\$1,474	4	4	0.519	C	B

Reclassification matrix.

\$-usage class \ multi-criteria class	A	B	C
A	28	12	0
B	12	37	11
C	0	11	89